

UrbanGuard: A User-Centered Real-Time Incident Monitoring System for Urban Safety

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Abstract—Urban emergencies such as road accidents, fires, thefts, and other safety incidents require timely communication between citizens and responsible authorities. However, incident information is often reported through fragmented communication channels, making it difficult for authorities to obtain complete and actionable information quickly. UrbanGuard is a user-centered mobile application concept designed to support real-time incident reporting, public safety awareness, and communication between citizens and emergency service personnel. The project followed an iterative human-computer interaction (HCI) process consisting of formative research, prototyping, and evaluation. Interviews with relevant stakeholders and surveys of general users were used to identify problems related to incident reporting, information management, and emergency communication. These findings informed low-fidelity and high-fidelity prototypes, which were refined through peer feedback, expert review, and think-aloud evaluation. The resulting prototype integrates incident reporting, real-time updates, evidence submission, safety guidance, and communication support within a unified interface. The study demonstrates how user-centered design can be applied to the development of a public-safety platform and identifies several opportunities for future work, including AI-assisted incident detection, GPS-based monitoring, multilingual support, offline reporting, and predictive safety analytics.

Index Terms—Urban safety, emergency response, incident reporting, human-computer interaction, user-centered design, mobile application, public safety

I. INTRODUCTION

Urban safety depends not only on the availability of emergency services but also on how quickly and accurately information about an incident reaches the appropriate responders. Road accidents, fires, thefts, medical emergencies, and other public-safety incidents can require immediate intervention, while delays in receiving relevant information can make response more difficult. Traditional reporting mechanisms often depend on telephone calls, direct communication, or manual reporting, which may result in incomplete information or delays in communication between citizens and responsible authorities.

The use of mobile and crowdsourcing technologies has been investigated as a potential means of improving emergency information collection. Jilani et al. proposed a crowdsourcing-based emergency event reporting framework in which nearby smartphone users could provide information such as incident location, incident type, and the number of victims to rescue services [1]. Their work demonstrates the potential of participatory sensing for collecting incident information from people near an emergency. Related research has also examined social-media-based emergency information and situational awareness [7]–[9].

Similarly, Huang et al. investigated mobile crowdsourcing for public-safety reporting and examined factors affecting people's decisions to report safety incidents [2]. Their experimental and interview-based work found that contextual factors, including location, can influence reporting decisions and subsequently informed the design of a location-based mechanism for encouraging safety reporting. Location-aware emergency applications have also been studied with respect to trust, privacy concerns, and user adoption [15].

Crowdsourcing has also been explored for emergency information management. CrowdEIM proposed using mobile social-media users to assist with emergency information-management tasks and incorporated formative studies into the design and evaluation of a mobile crowdsourcing tool [3]. Crisis informatics research further shows that social-media and crowdsourced information can contribute to situational awareness, while creating challenges related to filtering, prioritization, and information overload [10], [11].

More recent research has examined the reliability of crowd-sourced information in emergency management. Lelow and Marincioni investigated information obtained through emergency calls and social media during the 2018 Vaia Storm in Italy and reported differences in accuracy, reliability, and processing requirements between these information sources [4]. This highlights an important consideration for systems such as UrbanGuard: collecting more reports does not necessarily guarantee better emergency information. Reports must be appropriately structured, interpreted, and, where necessary, verified before supporting operational decisions. Similar concerns regarding information quality and the combination of social-media and authoritative data have been reported in disaster-management research [12], [13].

Social-media research also demonstrates the potential of real-time public information for emergency awareness. Vieweg et al. examined microblogging during natural hazards [7], while Kryvasheyev et al. found relationships between disaster-related social-media activity and physical disaster impacts [20]. These findings provide additional support for considering citizen-generated information as a complementary source for emergency awareness rather than as a replacement for authoritative response channels.

From an HCI perspective, the usability of a public-safety application is particularly important because users may interact with the system under time pressure or uncertainty. Nielsen's usability heuristics include principles such as visibility of system status, match between the system and the real world, user control and freedom, consistency and standards, and

recognition rather than recall [5]. These principles are directly relevant to UrbanGuard because users should be able to understand the state of an incident report, identify available actions, and access important information without unnecessary interaction.

The System Usability Scale (SUS), introduced by Brooke, provides a standardized method for obtaining a global measure of perceived usability through a short questionnaire [6]. SUS consists of ten items and can be used to complement qualitative usability evaluation. For an emergency-oriented system such as UrbanGuard, combining qualitative observations with a standardized usability measure can provide a more structured assessment of the interaction experience.

Although previous studies have investigated mobile crowdsourcing, emergency information management, and public-safety reporting independently, UrbanGuard focuses on bringing several related functions together within a user-centered mobile interface. The proposed system combines incident reporting, evidence submission, real-time incident information, public safety awareness, and communication support. The objective is not to replace existing emergency services, but to provide an additional structured communication layer between citizens and responsible authorities.

II. METHODOLOGY

UrbanGuard was developed using an iterative user-centered HCI methodology. The process consisted of three major stages: formative research, design and prototyping, and evaluation. Rather than treating these stages as completely independent activities, findings from each stage were used to refine the subsequent design.

A. Formative Research

The formative stage focused on understanding challenges associated with urban incident reporting and emergency communication. Two complementary research methods were used: semi-structured interviews and surveys.

Semi-structured interviews were conducted with individuals associated with public-safety and emergency-response activities, including police and traffic personnel. The interviews focused on incident reporting practices, communication with other departments, difficulties in identifying important information, and challenges associated with receiving or verifying incident reports. Open-ended questions allowed participants to describe their experiences and provide suggestions based on their existing workflows.

A survey was also used to collect responses from general users. The survey investigated people's experiences with emergency reporting, awareness of available reporting mechanisms, difficulties encountered during emergencies, and expectations regarding a mobile-based safety platform. Combining qualitative interviews with quantitative survey responses provided both detailed stakeholder perspectives and broader user-level observations.

The collected information was synthesized to identify recurring problems and requirements. The main issues identified

during this stage included difficulties in organizing incident information, delays in communication, limited public awareness, and the need for clearer coordination between citizens and relevant safety services.

B. Design and Prototyping

The design stage translated the findings from formative research into interface requirements and interaction concepts. Initial low-fidelity prototypes were created using paper sketches to explore navigation structures, screen layouts, and basic user flows. Low-fidelity prototyping allowed the team to modify the interface before investing significant effort in detailed visual design.

The revised low-fidelity prototypes for the official and general-user interfaces are shown in Figs. 1 and 2, respectively.

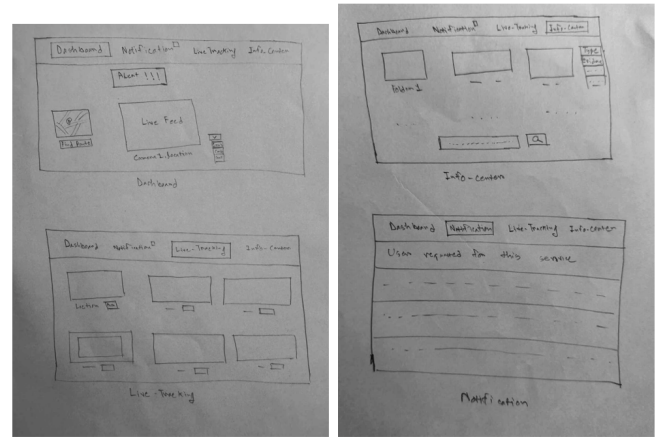


Fig. 1. Lo-Fi V2 sketches for official part after initial feedback.

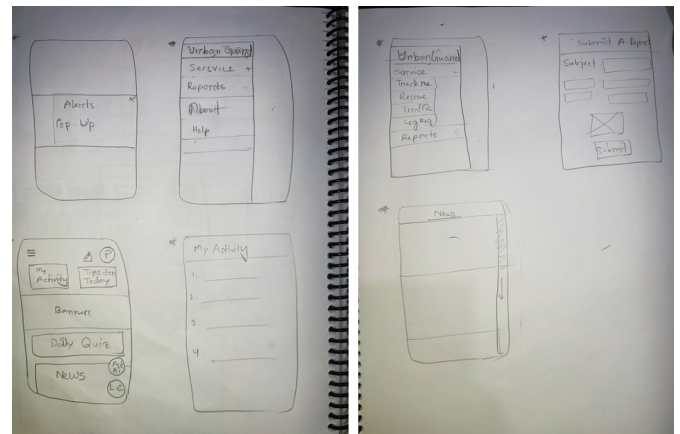


Fig. 2. Lo-Fi V2 sketches for user part after initial feedback.

The paper sketches were subsequently converted into digital prototypes using Figma. The prototype was organized around several related functions: incident reporting and evidence submission, real-time incident information, safety guidance and awareness, and communication or support services.

The corresponding high-fidelity prototypes are shown in Figs. 3–5.

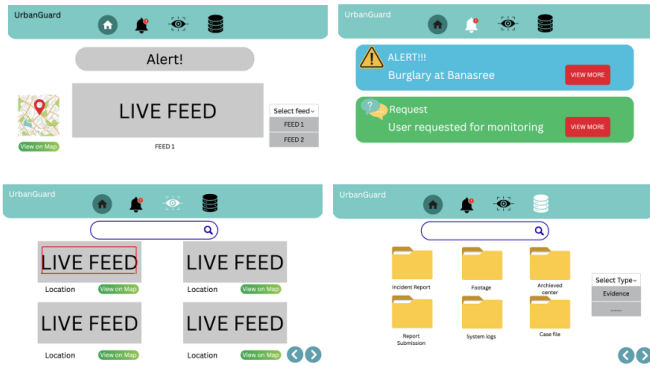


Fig. 3. Hi-Fi V2 prototypes for official part after feedback.

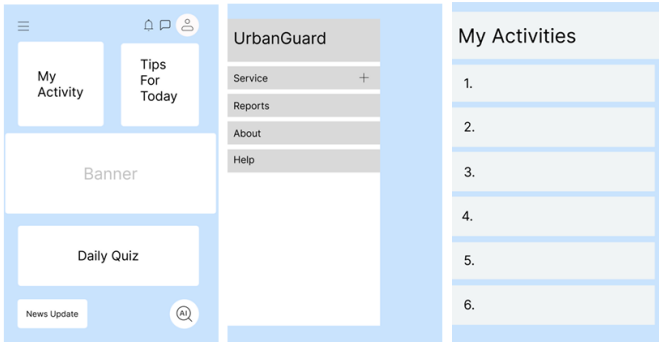


Fig. 4. Hi-Fi V2 prototypes for user part after feedback.

The design process was iterative. Peer feedback was used to identify problems with layout organization, icon interpretation, and the distinction between automated and human-assisted communication. Based on this feedback, subsequent prototype versions simplified navigation, improved the organization of information, and changed the general-user interface from landscape to portrait orientation to better suit mobile interaction.

The high-fidelity prototype incorporated more detailed visual components, interactive navigation, notifications, evidence submission, and dashboard elements. Additional attention was given to the visibility of important information, consistency between screens, and accessibility of interactive elements.

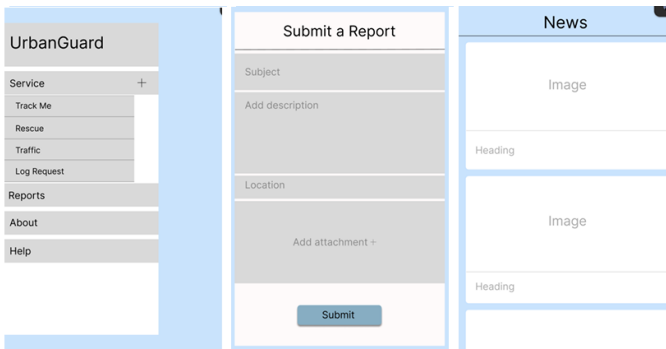


Fig. 5. Hi-Fi V2 prototypes for user part after feedback (2).

C. Evaluation

Evaluation was conducted using a combination of expert review and think-aloud testing. The prototype was presented to representative users and stakeholders, including police personnel, traffic officers, and students. Participants were asked to interact with important system functions while explaining their actions and difficulties.

The evaluation focused on whether users could understand the purpose of the interface, locate important information, complete reporting-related tasks, and interpret system feedback. The interface was also examined using established HCI usability principles, particularly visibility of system status, match between the system and the real world, user control, and help and documentation [5].

SUS was considered as a quantitative usability measure for the prototype. The original SUS instrument contains ten statements rated using a five-point response scale and produces an overall usability score [6]. In the present project, qualitative evaluation findings were primarily used to guide interface refinement.

The overall development progression from initial concepts to the refined prototype is summarized in Fig. 6.

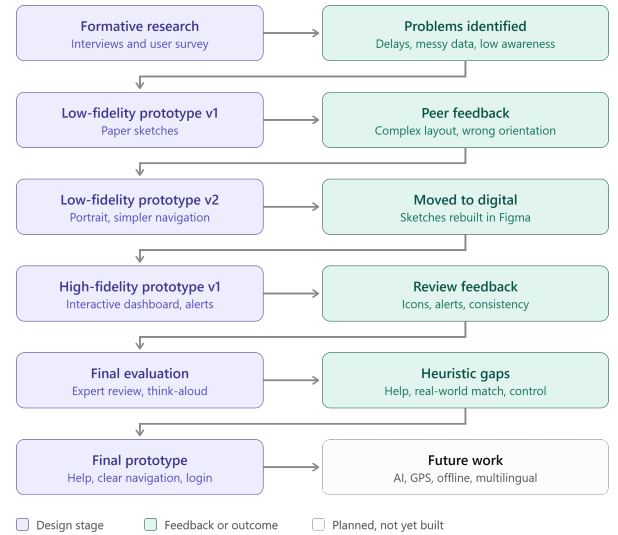


Fig. 6. Development progress of UrbanGuard.

III. URBANGUARD SYSTEM DESIGN

UrbanGuard was designed as a mobile platform connecting citizens with relevant emergency and public-safety services. The design emphasizes rapid access to important functions while attempting to avoid unnecessary interaction during stressful situations.

The central interaction flow allows a user to report an incident by providing relevant information, such as an incident category, description, location, and supporting evidence. Evidence may include photographs or other information that can help authorities understand the situation. The system is intended to organize submitted information so that it can be reviewed and acted

upon by authorized personnel. This approach is consistent with prior emergency-reporting systems that use mobile devices to collect structured incident information [1], [3].

A second component provides real-time incident information to users. This allows users to become aware of incidents in their surroundings and potentially avoid affected areas. The interface can present incident information through a dashboard or map-oriented view, depending on the specific implementation. Location-based emergency applications have previously been investigated in relation to user acceptance, trust, and privacy [15].

The system also includes safety guidance and awareness functions. These functions provide users with information intended to improve their understanding of appropriate responses to different emergency situations. Quizzes and guidance materials were incorporated into the prototype to encourage public engagement with safety information.

Another component provides support and communication functions. The prototype distinguishes between automated assistance and human support so that users can understand whether they are interacting with an automated system or a responsible service representative. This distinction was introduced following early prototype feedback.

Security and access control were also considered in the design. Sensitive operations and information should be accessible only to authorized users. The proposed system therefore includes authentication and role-based access concepts for operations involving sensitive incident information. Privacy and trust are particularly relevant for location-based emergency systems [15].

The complete conceptual workflow of UrbanGuard is illustrated in Fig. 7.

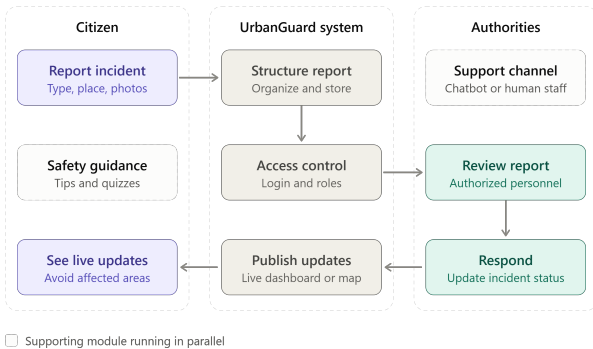


Fig. 7. Workflow of the UrbanGuard system.

IV. EVALUATION AND RESULTS

The development of UrbanGuard involved several rounds of refinement. The results of formative research first established the main problem areas, while subsequent prototype evaluations identified usability and interaction issues.

The formative research indicated that participants were concerned with delays in incident communication, difficulty organizing incident information, and limited awareness of

appropriate safety procedures. These observations informed the decision to combine reporting, real-time information, evidence submission, and safety guidance rather than treating incident reporting as an isolated function. Similar motivations for combining citizen-generated information with emergency-management workflows have been discussed in prior research [14], [17].

The first low-fidelity prototype provided a basic representation of the intended interaction flow. Peer feedback identified several usability concerns, including an overly complex visual organization, unclear differentiation between chatbot and human support, and an unsuitable orientation for the general-user interface. These findings resulted in a second low-fidelity version with simplified navigation, a portrait layout, and clearer communication pathways.

The transition to the high-fidelity prototype introduced more detailed interface components and interactive functionality. Evaluation of the first high-fidelity version identified issues involving icon clarity, message placement, visual consistency, and the visibility of important alerts. These findings were used to reorganize the dashboard and increase the prominence of safety-critical information.

The final evaluation identified usability issues corresponding to several established heuristic principles. In particular, the prototype required stronger help and documentation, clearer connections between interface concepts and real-world emergency workflows, and greater user control. These observations are consistent with established usability principles concerning help and documentation, match between the system and the real world, and user control and freedom [5].

Based on the evaluation findings, contextual assistance and clearer navigation were incorporated into the final prototype. Authentication was also introduced for security-sensitive operations. These modifications were intended to reduce confusion, improve task completion, and make the interface more appropriate for public-safety scenarios.

V. DISCUSSION

The development process demonstrates the importance of combining domain-specific stakeholder input with general user feedback when designing public-safety interfaces. Emergency applications have different requirements from conventional mobile applications because users may need to perform actions quickly while dealing with uncertainty or stress.

The formative research helped prevent the design from being limited to a simple incident-reporting interface. Instead, the prototype evolved toward a broader platform that includes reporting, evidence management, real-time information, awareness, and communication support. This direction is consistent with previous research showing that mobile crowdsourcing and social-media systems can contribute to emergency information collection and public-safety awareness [1]–[3], [7].

At the same time, the project highlights an important design challenge: collecting more information does not necessarily produce better emergency response. Crowd-sourced information can contain inaccurate, incomplete, or delayed reports [4],

[10]. Crisis-informatics research similarly emphasizes filtering, prioritization, verification, and the management of information overload [11], [19]. Therefore, a practical implementation of UrbanGuard would require mechanisms for verification, prioritization, and controlled information sharing.

Research on emergency information also suggests that combining citizen-generated information with authoritative sources can improve the interpretation of crisis-related information [13]. Similarly, PetaJakarta.org demonstrated the potential of crowdsourced data for disaster mapping while identifying implementation challenges [12]. These findings support UrbanGuard's future direction toward structured and verified incident information rather than unrestricted information aggregation.

The iterative prototype process also showed that relatively small interface decisions can have significant implications for emergency interaction. Clear icons, visible alerts, understandable terminology, predictable navigation, and distinguishable communication channels can reduce the effort required to interpret the interface. These observations support the use of established usability principles during the design of safety-critical interfaces [5].

The current work should therefore be understood as a user-centered prototype rather than a deployed emergency-response infrastructure. Actual deployment would require technical integration with authorized emergency services, appropriate security mechanisms, privacy protection, operational policies, and extensive real-world testing.

VI. FUTURE WORK

Several extensions can be explored in future iterations of UrbanGuard. First, AI-assisted incident detection could be incorporated to identify or assist in verifying events from submitted images or videos. Research on AI and multimodal social-media content indicates opportunities for automated extraction of disaster-related information while also highlighting challenges in reliability and operational use [16]. Such functionality could reduce the amount of manual screening required from authorized personnel, although automated detections would require careful validation before being used for operational decisions.

Second, GPS-based live mapping could be implemented to display nearby incidents and provide more precise location information to responders. Location-based functionality could also support routing and awareness of potentially affected areas. Prior work has examined geographic relationships between social-media information and authoritative disaster data [13].

Third, multilingual support could make the application accessible to users with different language backgrounds. An offline reporting mechanism could also be developed so that users can prepare reports when connectivity is unavailable and transmit them once a connection is restored.

Direct communication features, such as secure chat or voice communication with authorized emergency services, represent another possible extension. Such functionality would require appropriate authentication, routing, and escalation procedures.

Finally, aggregated incident data could be analyzed to identify temporal and geographical patterns. Long-term analysis could support planning and resource allocation by authorized authorities. Research has demonstrated that social-media activity can provide information related to disaster impacts and situational awareness [18], [20]. However, such analytics should incorporate appropriate privacy protections and should not expose sensitive personal or location information unnecessarily.

VII. CONCLUSION

UrbanGuard was developed as a user-centered approach to improving communication between citizens and public-safety services during urban incidents. The project combined formative interviews and surveys with iterative prototyping and usability evaluation to identify user needs and progressively refine the interface.

The resulting prototype integrates incident reporting, evidence submission, real-time incident information, safety awareness, and communication support within a unified mobile interface. Feedback obtained throughout the design process led to improvements in navigation, interface organization, alert visibility, communication clarity, and access to assistance.

The project also demonstrates that designing an emergency-oriented application requires more than implementing technical functionality. Information must be presented clearly, important actions must be easy to locate, and the system must account for the uncertainty and time pressure associated with emergency situations. Established HCI principles and usability evaluation methods therefore provide an important foundation for developing such systems.

UrbanGuard remains a prototype and requires further technical, operational, security, and real-world validation before deployment. Future work involving AI-assisted detection, GPS-based monitoring, multilingual and offline support, secure communication, and privacy-aware incident analytics could extend the system toward a more comprehensive urban safety platform.

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